

31338 Network Servers

Lab 3b Date, Time and NTP Synchronisation

Aims

1. Set the correct timezone and set the current time in Linux using an NTP server
2. Set up an NTP daemon so that the correct time is always maintained

Task 1: Set the time zone in Linux

Use the *date* command to identify the current UTC and local times (using the *-u* option for UTC, and no option for the local timezone). If the local time zone is incorrectly identified, change */etc/localtime* into a symbolic link to the correct file for our timezone under the directory */usr/share/zoneinfo*.

Try to set your system to a different timezone.

Task 2: Automatic Synchronisation with Chrony

Chrony is the system that maintains network time. It consists of both a server (daemon) and a client, called *chronyd* and *chronyc* respectively.

This task is a matter of setting up the *chronyd* daemon to run when the machine boots, to maintain the correct system time. Begin by see whether *chronyd* is already running using

```
systemctl status chronyd
```

Use *systemctl* to enable and start *chronyd*.

To verify that *chrony* is working, use the client program (*chronyc*) as follows. Document these commands in your journal.

- *chronyc sources*
- *chronyc tracking*
- *chronyc add server time.uts.edu.au*

To test that it will synchronise, let us change the system date, e.g., 01/01/2000.

Then just run *date* with no arguments to check the date. Does it reset to the current date/time? Why or why not?

Restart *chronyd* and run the *date* command. What happens now? Why?

Record notes in your learning journal about what happens and why. Also look at the configuration file */etc/chrony.conf*

Task 3: Windows Server Time Configuration

Windows can also automatically update the time and date via NTP. In *Server Manager* -> *Local Server*, click on the *time zone* shown to open the Date and Time window. Go to the Internet Time tab.

You can then change the settings to choose your internet time server. The default is `time.microsoft.com` - try changing this to: `time.uts.edu.au`.

You can also use the PowerShell commands:

```
w32tm /query /source
```

```
w32tm /resync
```